

Two errors, not one: the blade tones are bent by up to 8 dB below 1 kHz, and under them sits a broadband error of ± 1.3 dB (2.7 dB worst) that no band is exempt from. **The arc is not qualified to a standard anywhere yet.** For these runs the arc was laid flat (horizontal), with the propeller axis vertical. All microphones then sit at the same height in the propeller plane, so a perfect setup reads the same level everywhere: a circle. Twelve runs (prop7-18, 1-2 Sept 2026, PWM 1900); what is left after each microphone's own offset is removed is the room.

remedies report: docs/arc-validation-remedies.pdf



Fig. 2. One fit over keys dp1-baseline-horizontal-prop7...prop18, PWM 1900, prop11-14 mirrored. Rows aligned: mic row i sat at spot row i. 0° mic never moved, so its M and the 0° P are known only as a sum. Third-octave centre, Hz.

49°0'	0.34	0.15	0.74	0.75	0.51	0.1	0.4	0.10	0.3	0.4	0.4	0.3	0.2
47°0'	0.1	0.1	0.19	0.2	0.1	0.6	0.6	0.5	0.5	0.2	0.2	0.1	0.3
45°0'	0.5	0.2	0.19	0.2	0.1	0.6	0.4	0.7	0.0	0.7	0.2	0.4	0.1
36°0'	0.7	0.4	0.1	0.1	0.1	0.2	0.1	0.1	0.0	0.1	0.1	0.1	0.1
0°0'	0.1	0.08	0.74	0.8	0.36	0.2	0.4	0.1	0.3	0.7	0.3	0.1	0.1
18°0'	0.1	0.08	0.91	0.9	0.40	0.1	0.1	0.1	0.2	0.6	0.0	0.3	0.2
36°0'	0.7	0.1	0.19	0.2	0.4	0.6	0.3	0.1	0.5	0.1	0.5	0.7	0.6
18°0'	0.6	0.04	0.1	0.1	0.1	0.1	0.1	0.1	0.2	0.1	0.1	0.0	0.3
54°0'	0.4	0.0	0.1	0.1	0.5	0.7	0.1	0.3	0.8	0.1	0.1	0.0	0.1
72°0'	0.1	0.1	0.6	0.4	0.1	0.0	0.4	0.7	0.1	0.5	0.3	0.1	0.1
90°0'	0.5	0.1	0.2	0.1	0.5	0.1	0.2	0.1	0.3	0.6	0.4	0.1	0.0

RUN	MIC THERE	MEASURED	GAIN	P ROOM	M MIC	MODEL	LEFT
prop18	811-1896	59.1	62.3	-4.6	+2.5	60.2	-1.1
prop12 (180°)	810-8901	58.6	62.2	-4.6	+1.4	59.0	-0.4

+54° at 800 Hz: prop8 with 810-8901 reads +5.6 re gain = P +4.7 + M +1.4;
prop18 with 811-1892 reads +3.1 = +4.7 - 4.2. Two mics, one room value.